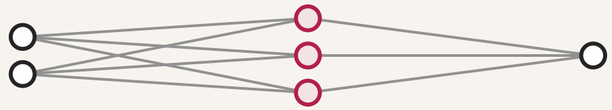
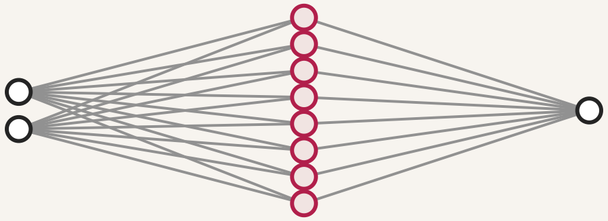
A neural network has no single shape.

Shallow / narrow



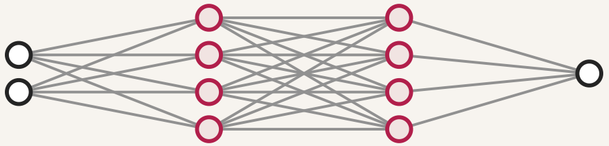
2 → 3 → 1

Shallow / wide



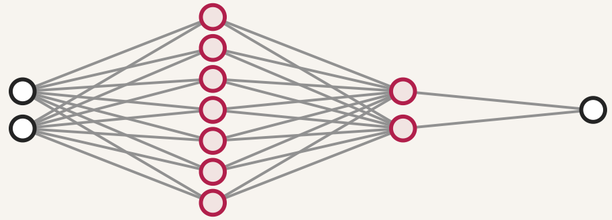
2 → 8 → 1

Two hidden layers



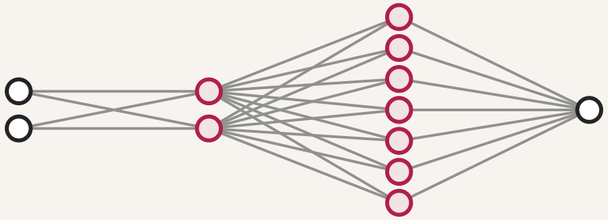
2 → 4 → 4 → 1

A bottleneck



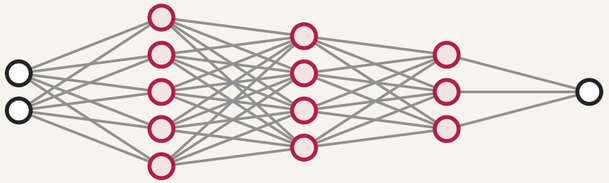
2 → 7 → 2 → 1

An expansion



2 → 2 → 7 → 1

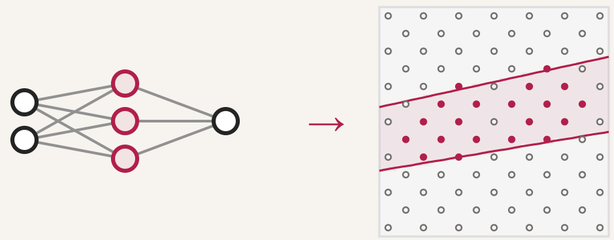
Three hidden layers



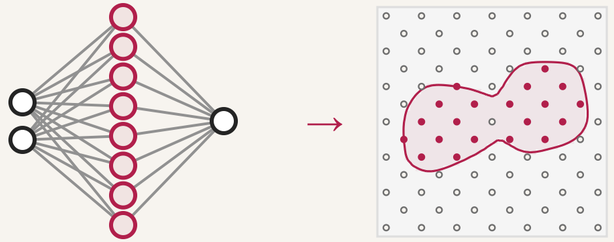
2 → 5 → 4 → 3 → 1

Different networks learn different boundaries.

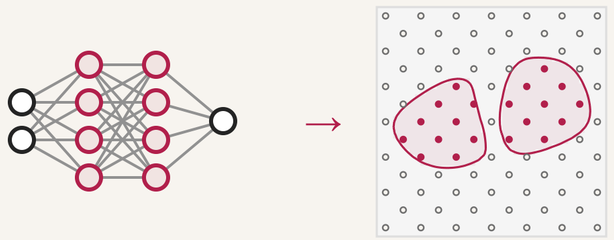
2 → 3 → 1



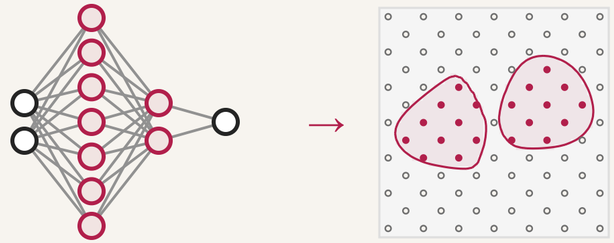
2 → 8 → 1



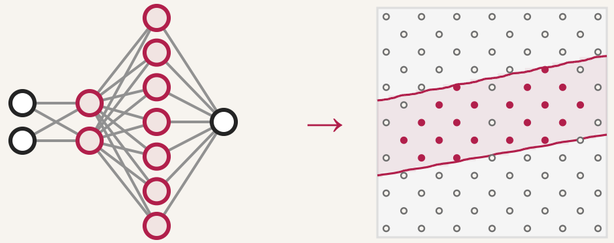
2 → 4 → 4 → 1



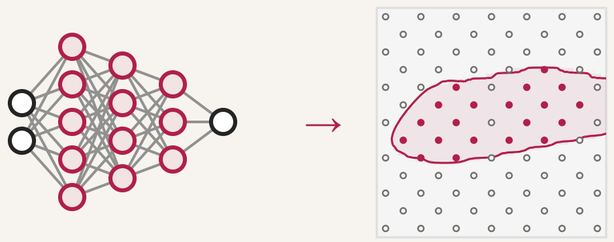
2 → 7 → 2 → 1



2 → 2 → 7 → 1



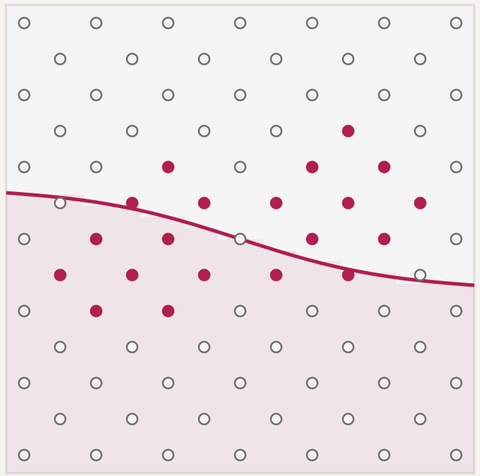
2 → 5 → 4 → 3 → 1



Same synthetic data. Actual fitted scores. Architecture alone does not determine the boundary.

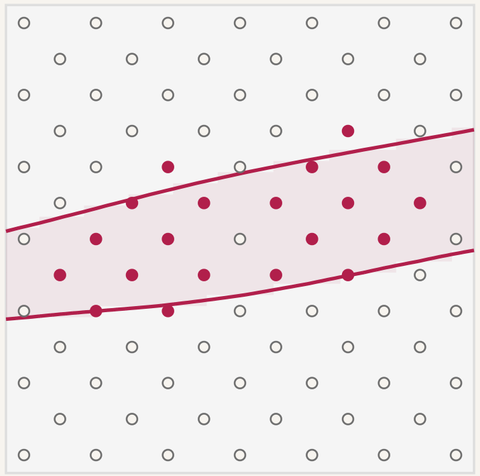
Training changes the function, not the wiring.

Before training



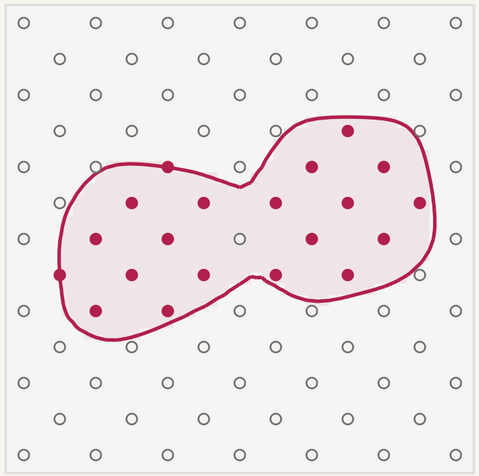
Mean training BCE = 0.7370

After 120 updates



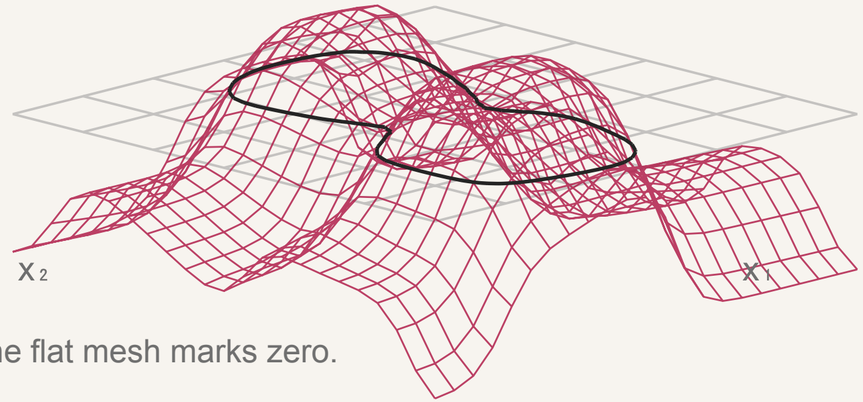
Mean training BCE = 0.2807

After 600 updates

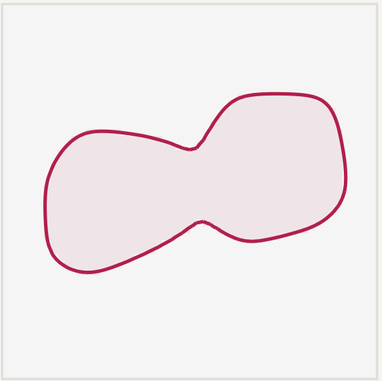


Mean training BCE = 0.0864

A boundary is a level set of a function.



Height = $z\theta(x_1, x_2)$. The flat mesh marks zero.



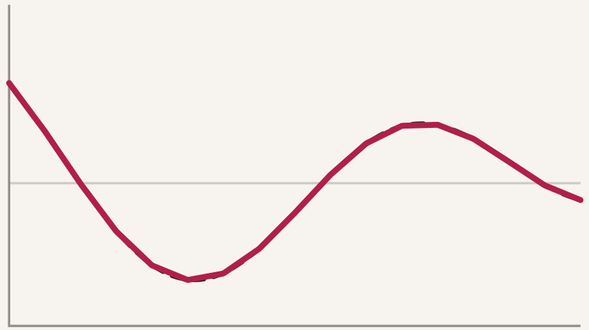
$$z\theta : \mathbb{R}^n \to \mathbb{R}$$

$$p = \text{sigmoid}(z\theta)$$

$$\hat{y} = 1[z\theta \geq 0]$$

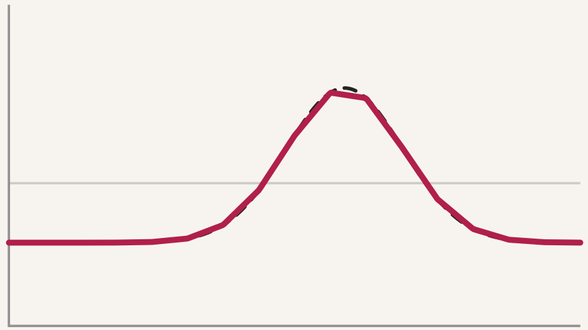
Different tasks ask for different functions.

Smooth oscillation



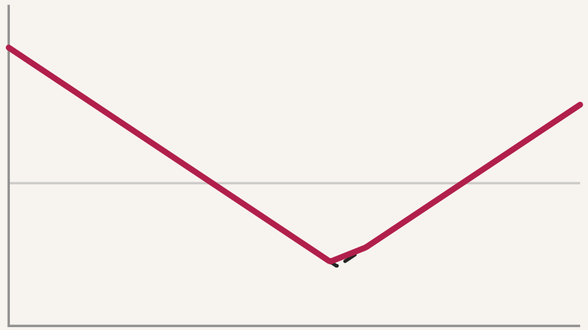
$$0.65\ \sin(2x) + 0.15x^2 - 0.25$$

A localized hill



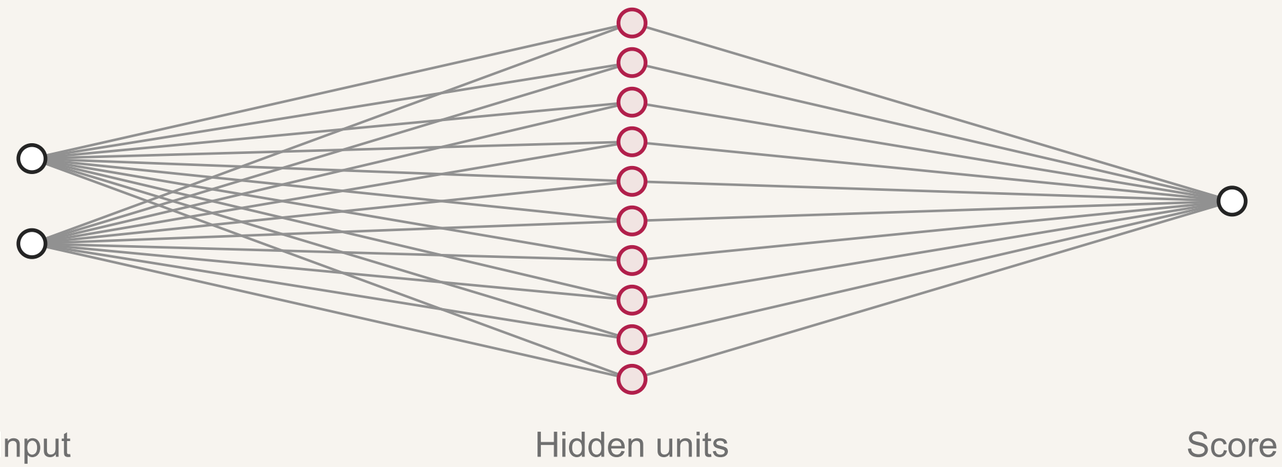
$$1.3\ \exp(-3(x - 0.35)^2) - 0.5$$

A continuous corner



$$0.8\ |x - 0.3| - 0.7$$

Single Hidden Layer NN: Universal Approximator



More units.
Not necessarily more layers.

$$f\theta(x) = b + \sum_j v_j \, g(a_j^T x + \beta_j)$$

Sufficient width + a suitable activation.

Universal does not mean exact—or effortless.

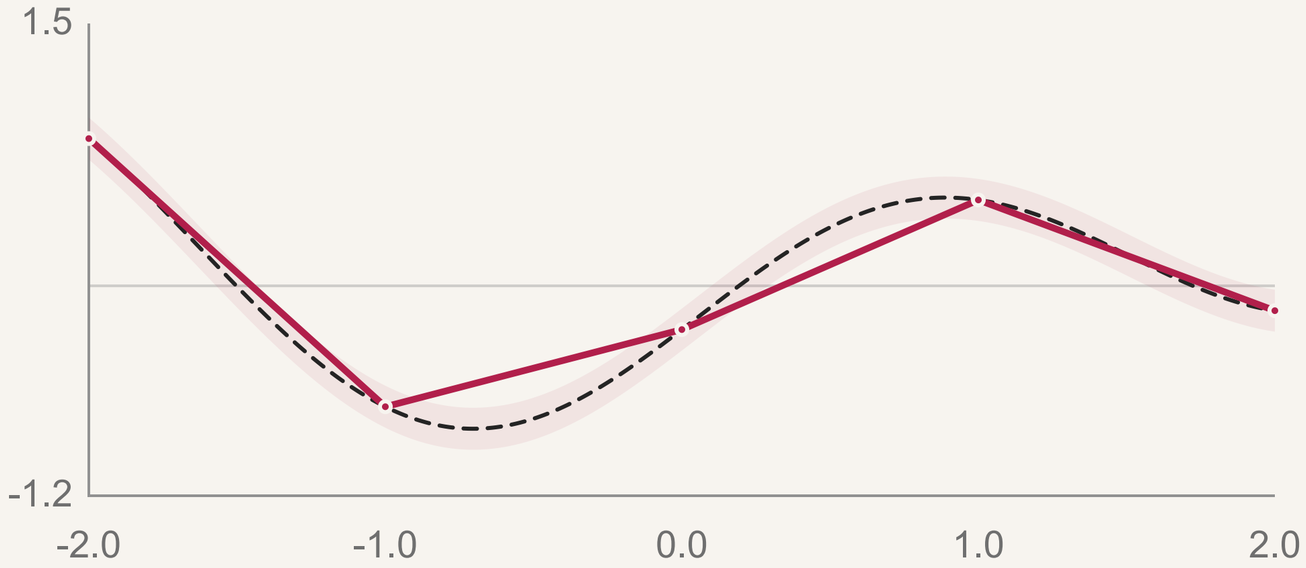
Continuous f on compact $K \subset \mathbb{R}^n$; for every $\varepsilon > 0$:

$$\exists\ m, \theta \quad \text{such that} \quad \max_{x \in K} |f(x) - f\theta(x)| < \varepsilon$$

One hidden layer. Suitable nonlinear activation. Enough units and suitable parameters.

For example: sigmoid or ReLU hidden units, with a linear readout.

More hidden units can refine the approximation.



Dashed: target. Solid: network. Band: ±ε around the target.

Target function

Smooth oscillation

Hidden units m 4

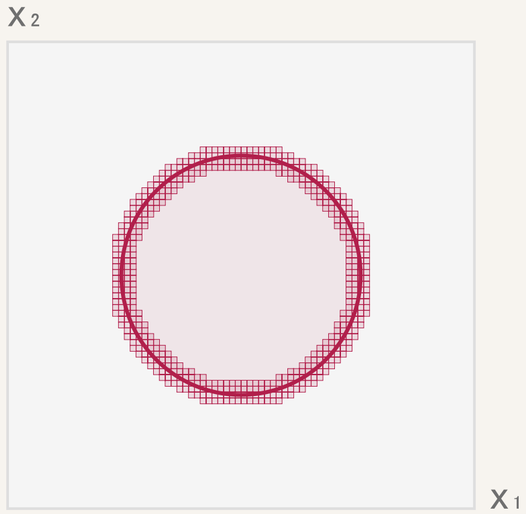
Tolerance ε 0.12

0.2914

Sampled max error ≥ ε = 0.12. 4 hidden ReLU units.

$$b + \sum_j v_j \, \mathrm{ReLU}(x - t_j)$$

Score accuracy protects labels away from zero.



$$\| f - f\theta \|_\infty < \varepsilon$$

Same sign wherever |f(x)| > ε.

Near the boundary, a small score error can still change the class.

The theorem leaves the design work to us.

01

Architecture

Depth and connectivity

02

Width

How many hidden units?

03

Activation

Which nonlinear features?

04

Loss

What counts as error?

05

Optimizer

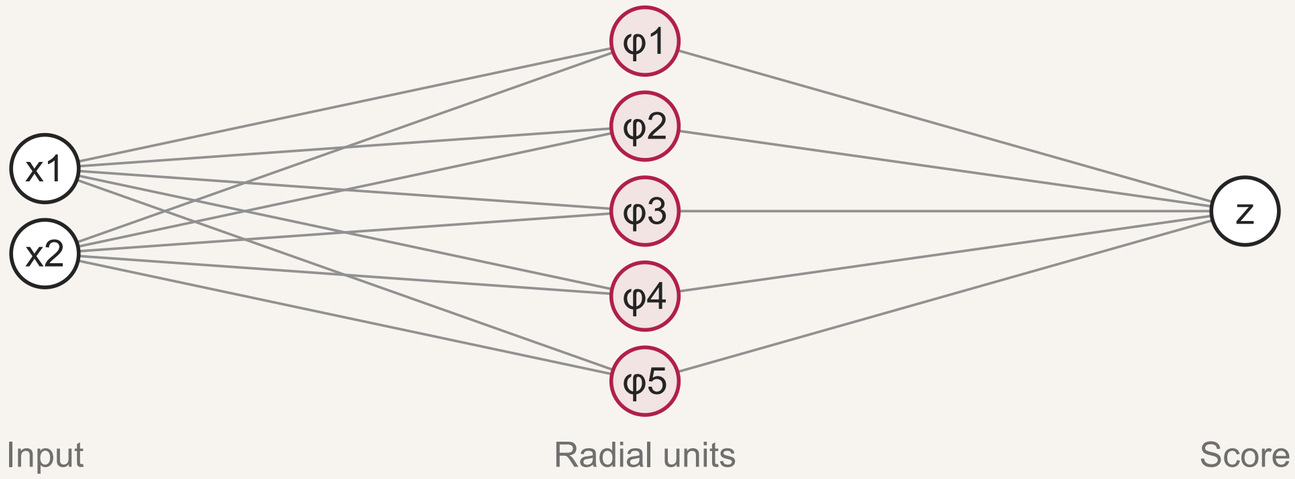
How do parameters move?

06

Generalization

What works on new data?

An RBF network chooses radial features.



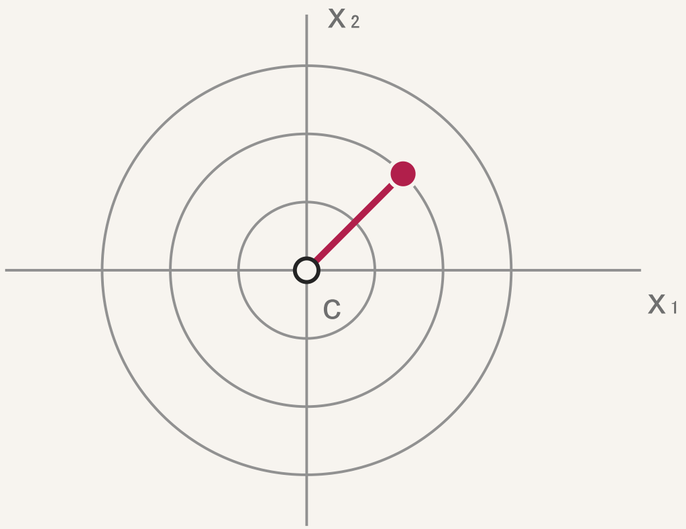
One hidden basis layer.

$$\varphi_j(x) = \psi(\|x - c_j\|, \sigma_j)$$

$$z = b + \sum_j w_j \varphi_j(x)$$

The classic RBF architecture; extended and deep variants also exist.

Radial means “distance, not direction”.



Distance r 1.00

Direction, degrees 45

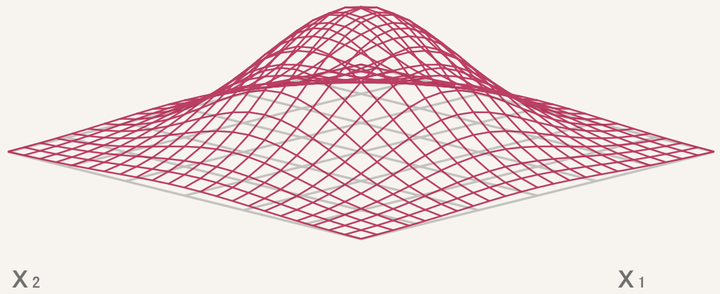
$$r = \sqrt{\sum_k (x_k - c_k)^2}$$

0.607

Gaussian response, σ = 1.
Use meaningfully scaled input features.

A bell and a bowl can both be radial.

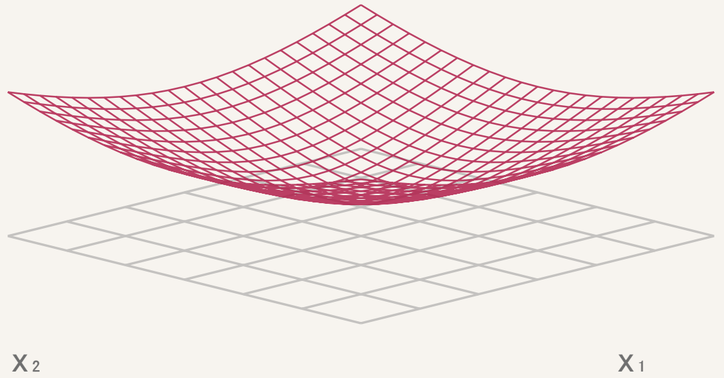
Gaussian



$$\psi(r,\sigma) = \exp(-r^2 / 2\sigma^2)$$

Localized: tends to zero far away.

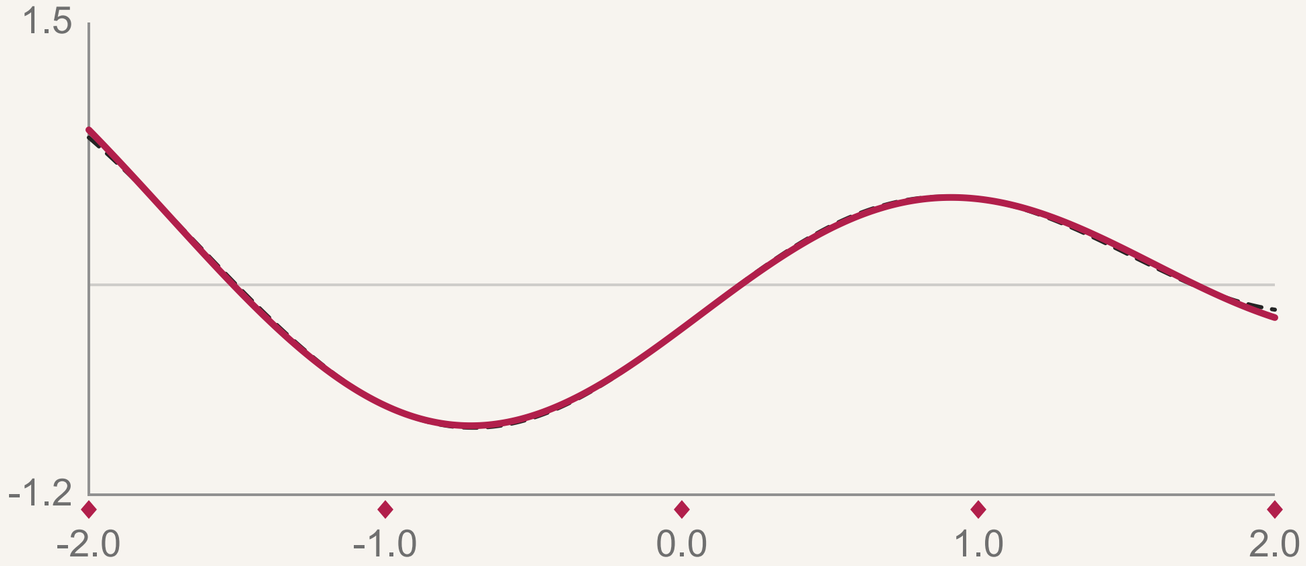
Multiquadric



$$\psi(r,\sigma) = \sqrt{(1 + r^2 / \sigma^2)}$$

Not localized: grows with distance.

Gaussian bases can approximate a whole function.



Dashed: target. Solid: RBF network. One hidden layer throughout.

Gaussian units m 5

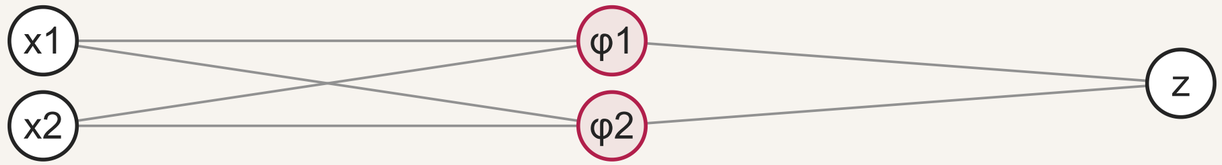
$$f\theta(x) = b + \sum_j w_j \, e^{-\|x-c_j\|^2/(2\sigma^2)}$$

0.0453

Sampled max error. 5 centers; σ = 1.100. Readout refitted to the target.

Gaussian RBF universality: Park & Sandberg, 1991.

Trace one input through a complete classifier.



Input Radial units Score

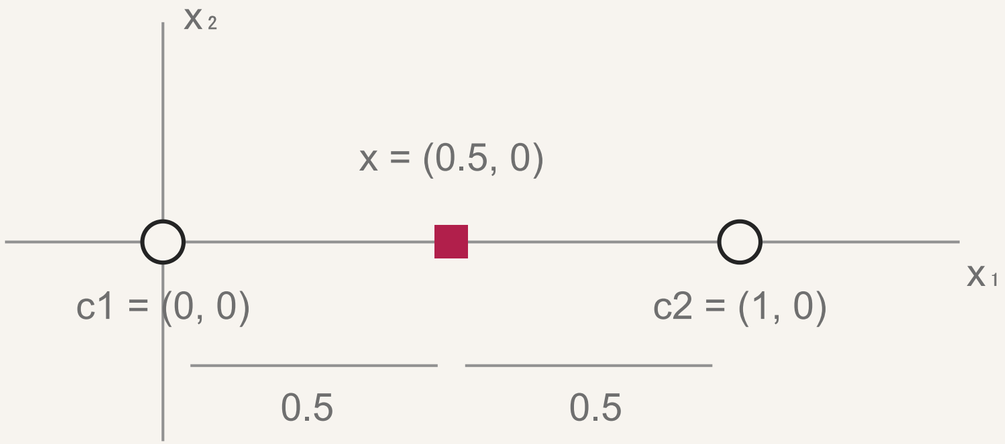
Synthetic walkthrough: x = (0.5, 0), label y = 1.

$$c_1 = (0, 0)$$
$$c_2 = (1, 0)$$
$$\sigma_1 = \sigma_2 = 0.75$$

$$w = [1.2, -0.8]$$
$$b = -0.1$$

Two units make the arithmetic visible—not a universal fixed-width claim.

Forward 1: measure squared distances.

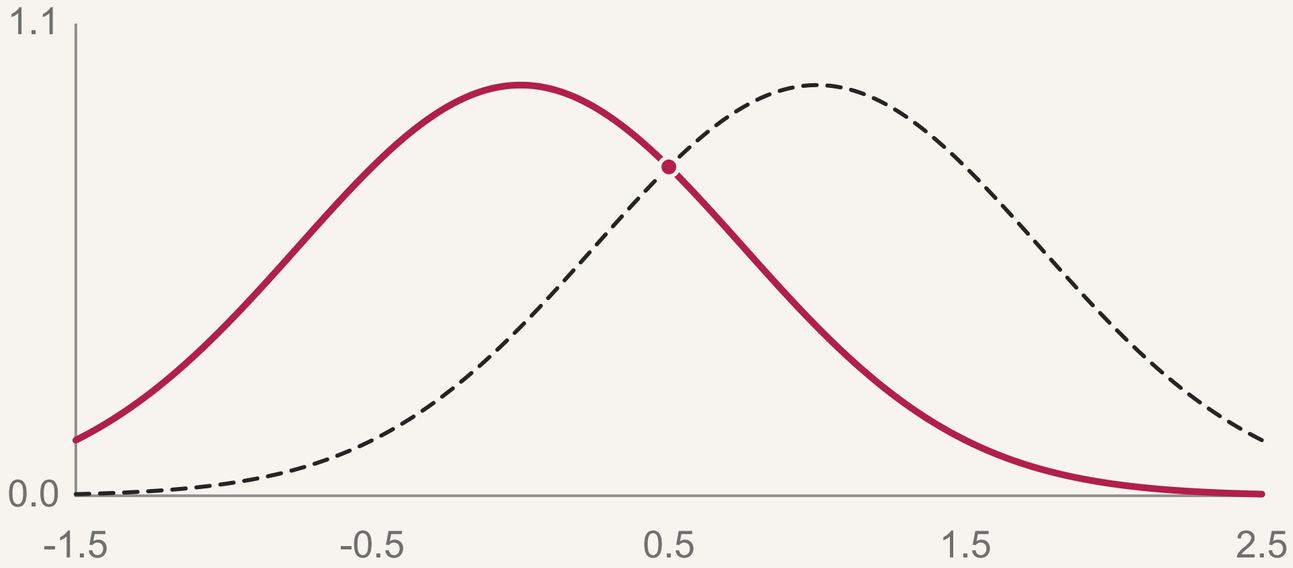


$$r_j^2 = \sum_k (x_k - c_{jk})^2$$

$r_1^2 = (0.5 - 0)^2 + 0^2$
$= 0.250$

$r_2^2 = (0.5 - 1)^2 + 0^2$
$= 0.250$

Forward 2: turn distance into a response.



A cross-section at $x_2 = 0$.

$$\varphi_j = \exp(-r_j^2 / 2\sigma_j^2)$$

```
exp(-0.25 / 1.125)
```

$$\varphi_1 = \varphi_2 = 0.801$$

Equal distance and equal width → equal response.

Forward 3: add contributions, then classify.

Signed contributions
`0.961, -0.641`

→

Add b = −0.1
z = 0.220

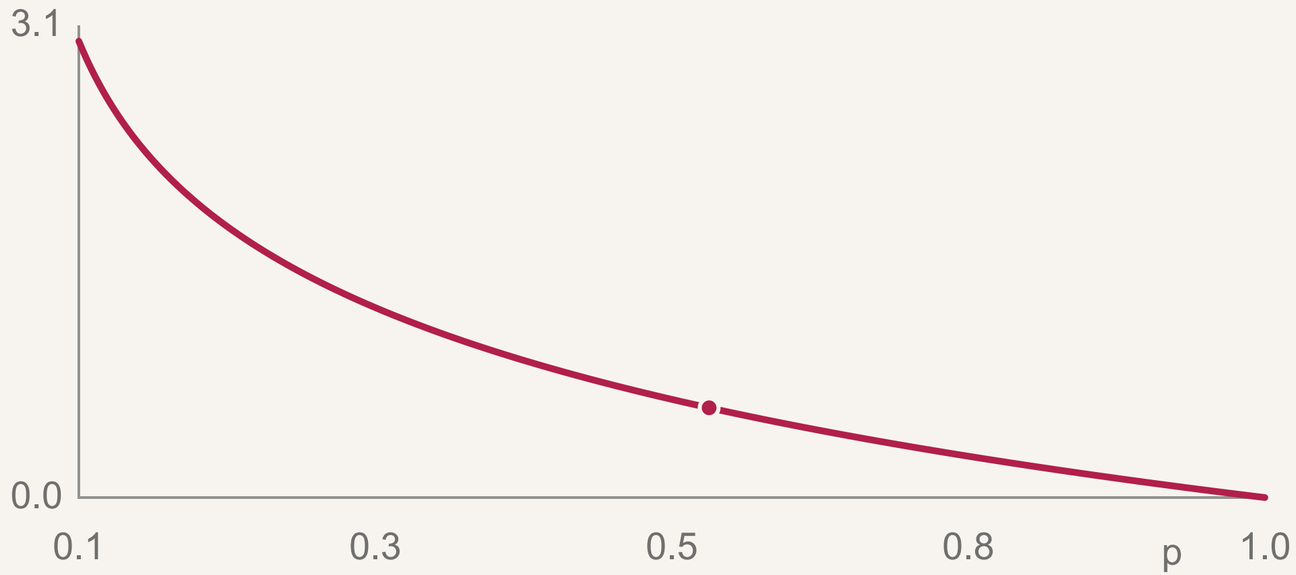
→

Sigmoid
p = 0.555

$$z = b + \sum_j w_j \varphi_j \qquad p = 1 / (1 + e^{-z})$$

p ≥ 0.5 → predict class 1.

Forward 4: even a correct label has a loss.



$$L = -y \log p - (1-y) \log(1-p)$$

$$y = 1 \rightarrow L = -\log p$$

$$L = 0.589$$

Correct class, but only moderate confidence.

Backward 1: the score receives one error signal.

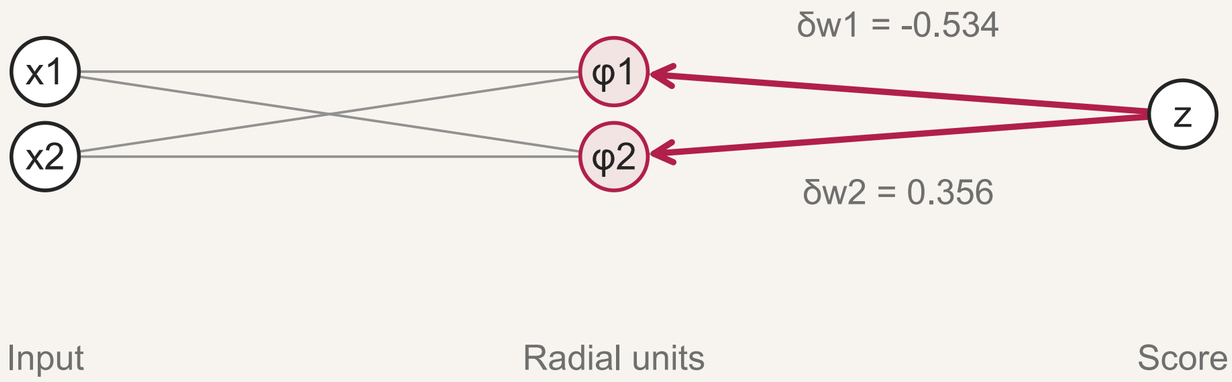
$$\partial L/\partial p = -y/p + (1-y)/(1-p)$$

$$\partial p/\partial z = p(1-p)$$

$$\delta \equiv \partial L/\partial z = p - y = -0.445$$

Negative δ: increasing this example’s score reduces its loss.

Backward 2: update the readout weights.



$$\partial L/\partial w_j = \delta \varphi_j$$

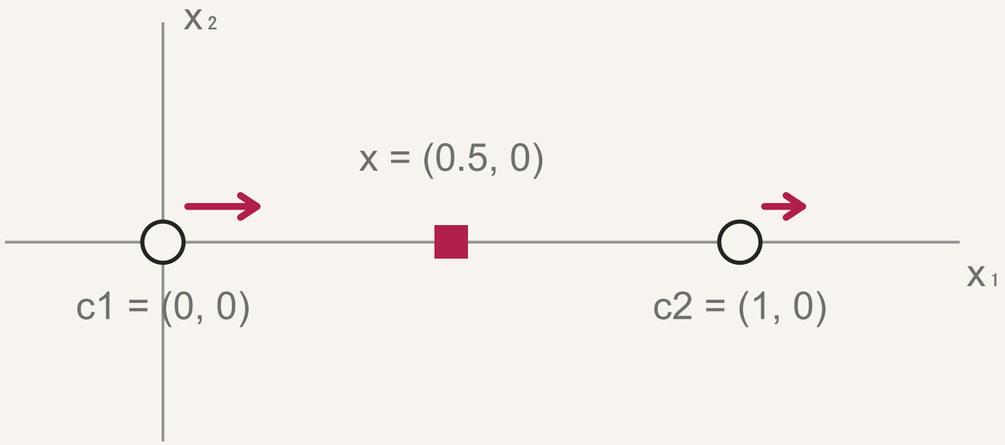
$$\partial L/\partial b = \delta$$

```
∂L/∂w = [-0.356, -0.356]
```

```
∂L/∂b = -0.445
```

Each gradient uses the cached forward response.

Backward 3: centers can move in feature space.



Arrows show $-\partial L/\partial c_j$, enlarged for visibility.

$$\partial L/\partial c_j = \delta w_j \varphi_j (x - c_j) / \sigma_j^2$$

$$\partial L/\partial c_1 = [-0.380,\ 0.000]$$

$$\partial L/\partial c_2 = [-0.253,\ 0.000]$$

The sign of w_j matters: not every center moves toward x.

Backward 4: learn widths without going negative.



Dashed: original. Solid: after $\eta = 0.2$.
Positive expert broadens; negative expert narrows.

$$\sigma_j = \exp(\rho_j)$$

$$\partial L/\partial \rho_j = \delta w_j \varphi_j \, r_j^2 / \sigma_j^2$$

```
∂L/∂ρ = [-0.190, 0.127]
```

$$\rho_j \leftarrow \rho_j - \eta \, \partial L/\partial \rho_j$$

σ_j stays positive.

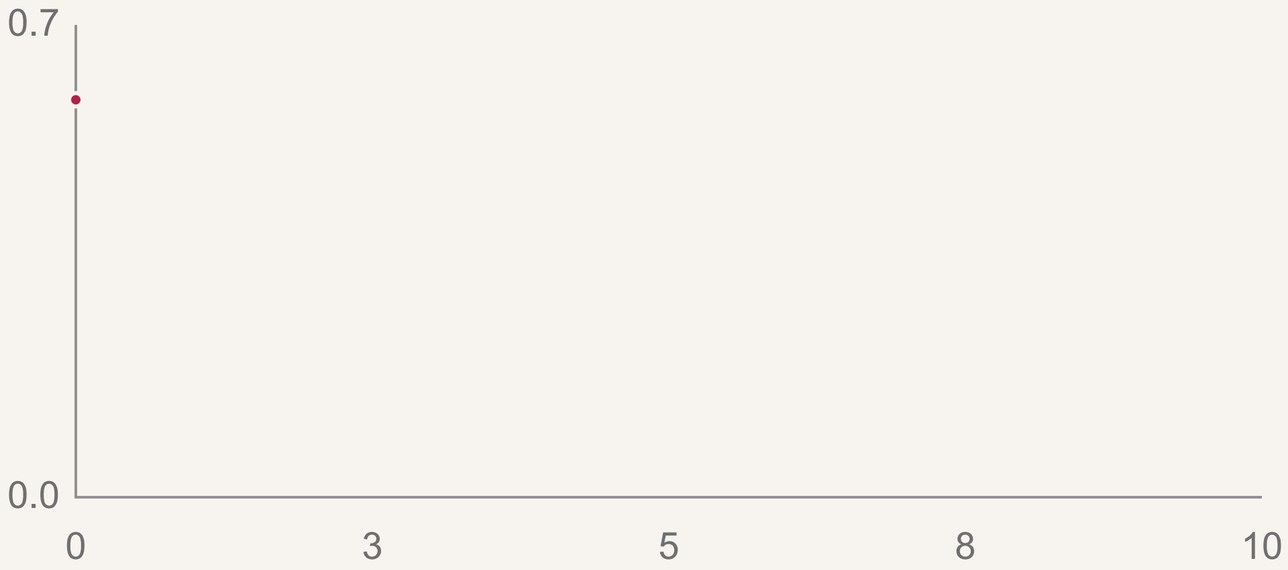
A different radial function changes its derivative.

Radial family	$\partial\varphi_j / \partial c_j$	$\partial\varphi_j / \partial\rho_j$
Gaussian	$\varphi_j(x - c_j) / \sigma_j^2$	$\varphi_j r_j^2 / \sigma_j^2$
Multiquadric	$-(x - c_j) / (\sigma_j^2 \varphi_j)$	$-r_j^2 / (\sigma_j^2 \varphi_j)$

$$\partial L/\partial(\text{geometry}) = \delta w_j \times \partial\varphi_j/\partial(\text{geometry})$$

Multiquadric convention: $\varphi_j = \sqrt{(1 + r_j^2/\sigma_j^2)}$. $\rho_j = \log \sigma_j$.

Apply all updates from the same forward pass.



Same one example only—not evidence of generalization.

$c_{11} = 0.000$ $c_{21} = 1.000$ $\sigma = [0.750, 0.750]$

Learning rate η 0.20

$$\theta \leftarrow \theta - \eta \nabla \theta L$$

Take one step | Reset

p = 0.555

Loss = 0.5891 · Update 0 / 30

Start from the worked example. Each click recomputes the entire gradient.

Choose whether to train just weights—or geometry.

Fixed centers and widths

Learn w and b.

$$\nabla_w L, \quad \partial L/\partial b$$

Logistic readout: convex objective.

End-to-end

Learn w, b, c and ρ.

$$\nabla_w L, \quad \partial L/\partial b, \quad \nabla_c L, \quad \nabla_\rho L$$

Moving geometry: generally nonconvex.

Architecture chooses the family. Training chooses fθ.

Architecture
Which functions?

→

Forward pass
Which prediction?

→

Backward pass
Which change?

Universality says a solution can exist. Learning still has to find a useful one.

Sources: Cybenko (1989) · Park & Sandberg (1991) · Universal approximation overview · Radial families